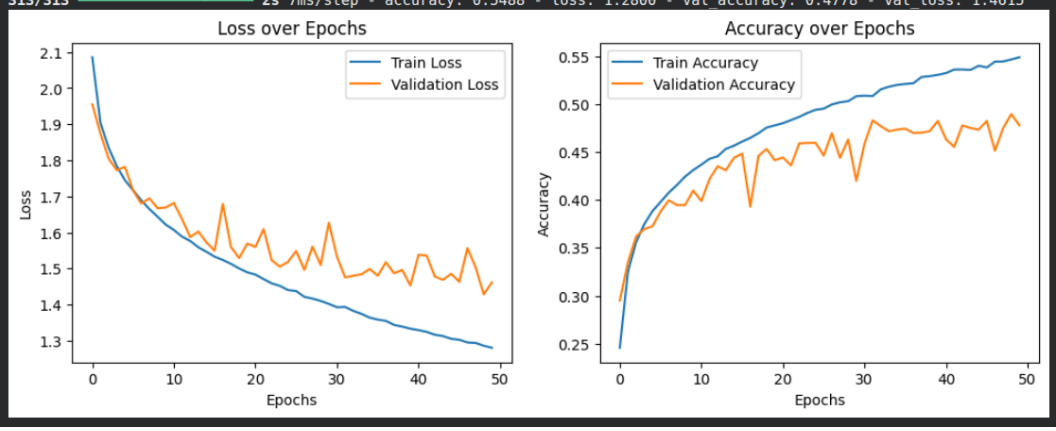
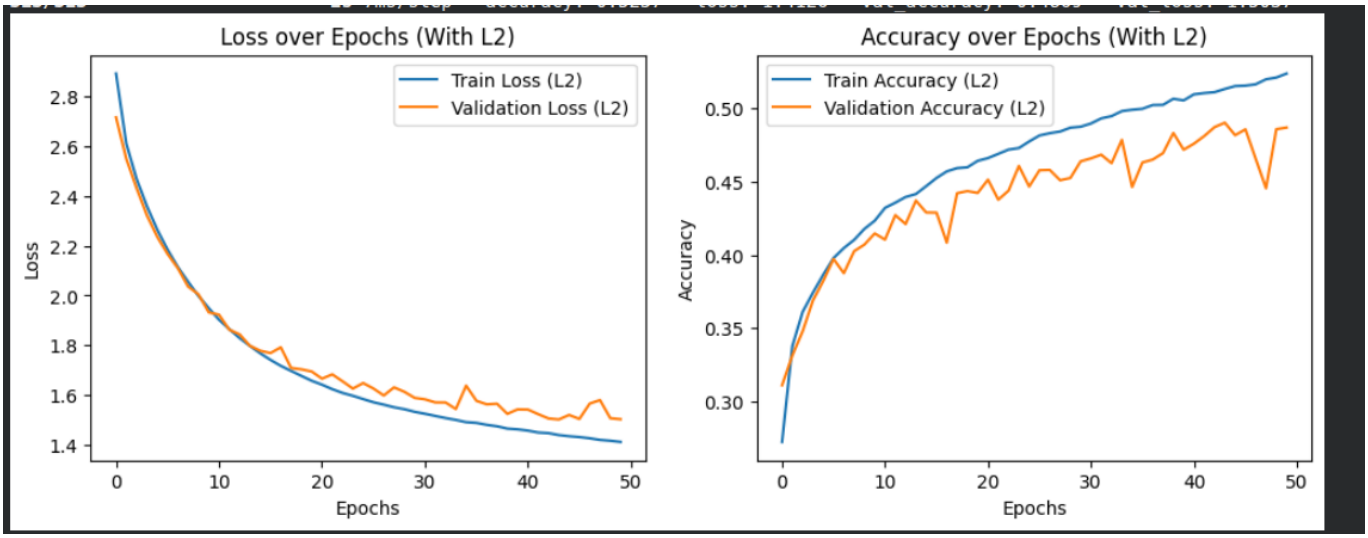
Q4:

- 4- Train the designed architecture using Mini-batch Stochastic Gradient Descent (SGD) with a batch size of 128, a learning rate of 0.01, for 50 epochs.

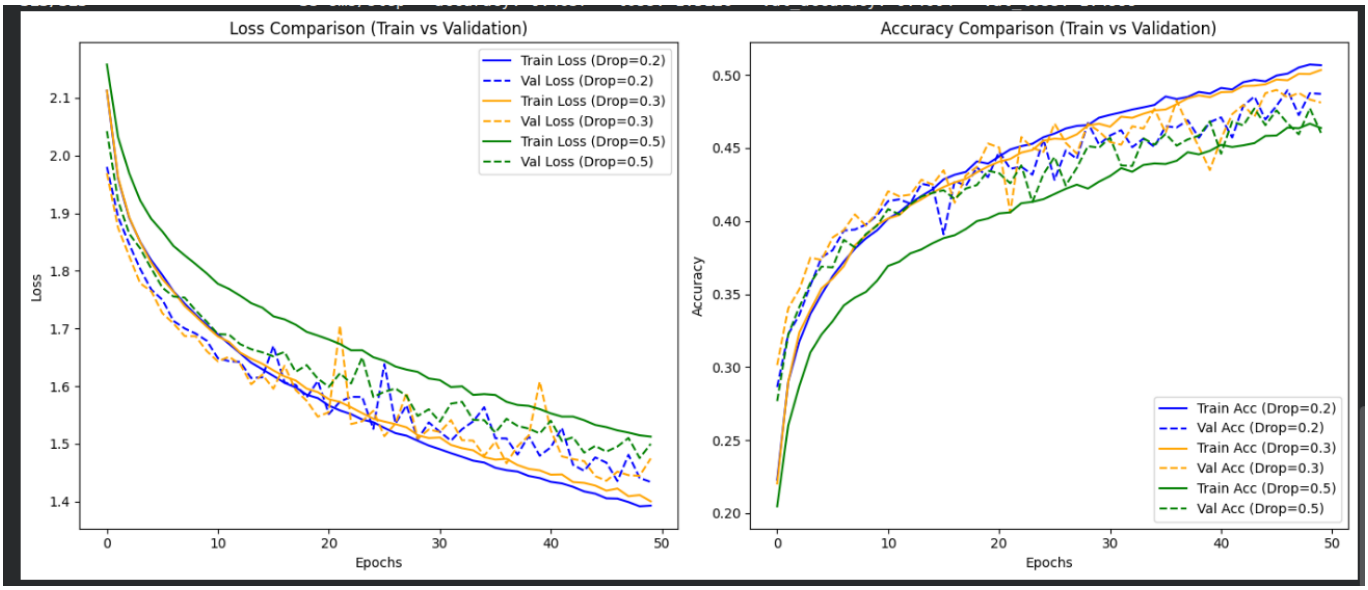
```
Epoch 45/50
313/313 ━━━━━━━━━━━━━━━━━━━━ 2s 7ms/step - accuracy: 0.5400 - loss: 1.3050 - val_accuracy: 0.4733 - val_loss: 1.4853
Epoch 46/50
313/313 ━━━━━━━━━━━━━━━━━━━━ 3s 7ms/step - accuracy: 0.5382 - loss: 1.3023 - val_accuracy: 0.4823 - val_loss: 1.4630
Epoch 47/50
313/313 ━━━━━━━━━━━━━━━━━━━━ 2s 7ms/step - accuracy: 0.5443 - loss: 1.2948 - val_accuracy: 0.4513 - val_loss: 1.5563
Epoch 48/50
313/313 ━━━━━━━━━━━━━━━━━━━━ 3s 9ms/step - accuracy: 0.5444 - loss: 1.2934 - val_accuracy: 0.4748 - val_loss: 1.5031
Epoch 49/50
313/313 ━━━━━━━━━━━━━━━━━━━━ 2s 7ms/step - accuracy: 0.5465 - loss: 1.2857 - val_accuracy: 0.4895 - val_loss: 1.4284
Epoch 50/50
313/313 ━━━━━━━━━━━━━━━━━━━━ 2s 7ms/step - accuracy: 0.5488 - loss: 1.2800 - val_accuracy: 0.4778 - val_loss: 1.4615
```



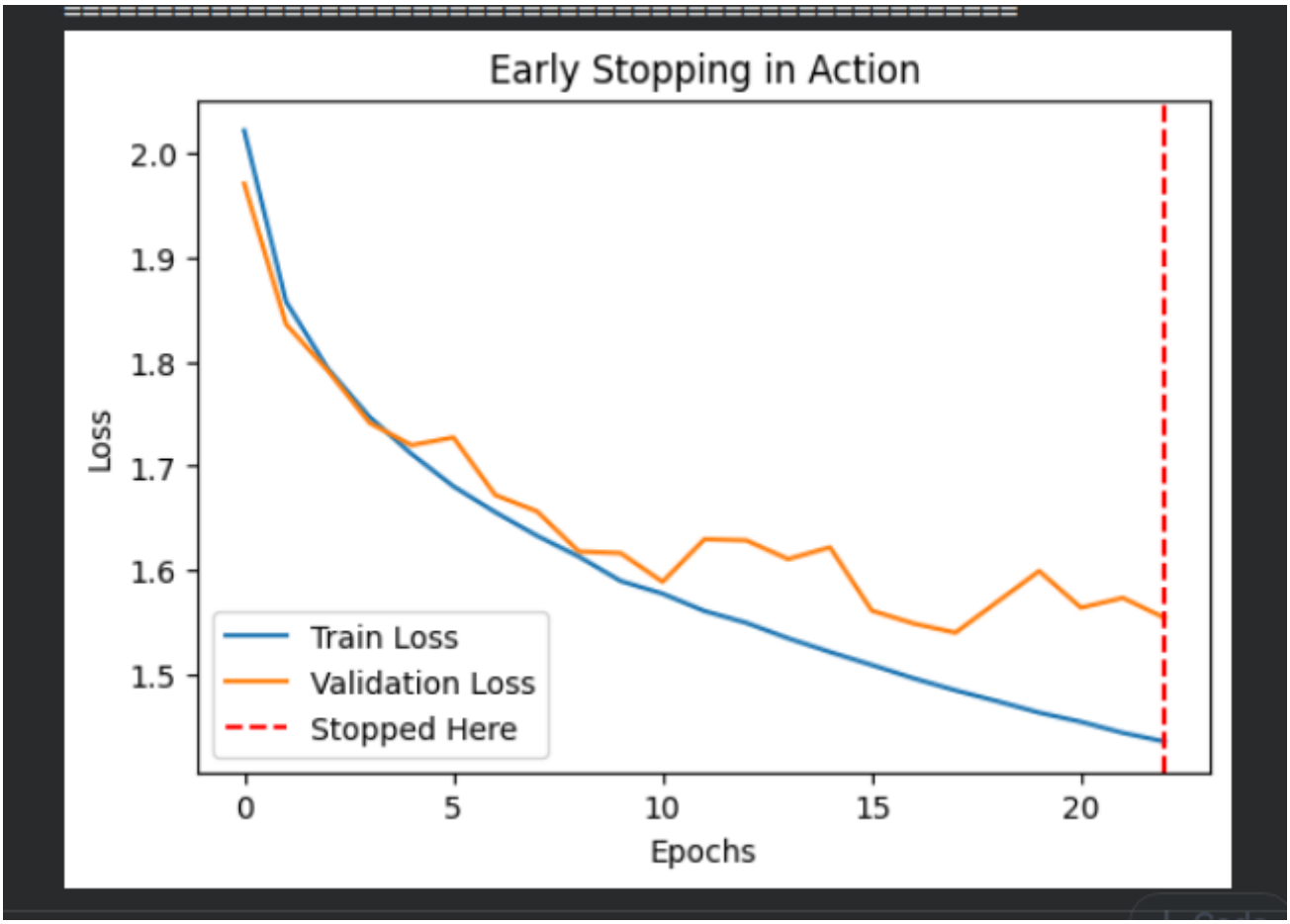
Q5 : Add L2 norm regularization to the second fully connected layer



Q6 : Add a dropout layer with rates of 0.2, 0.3, and 0.5

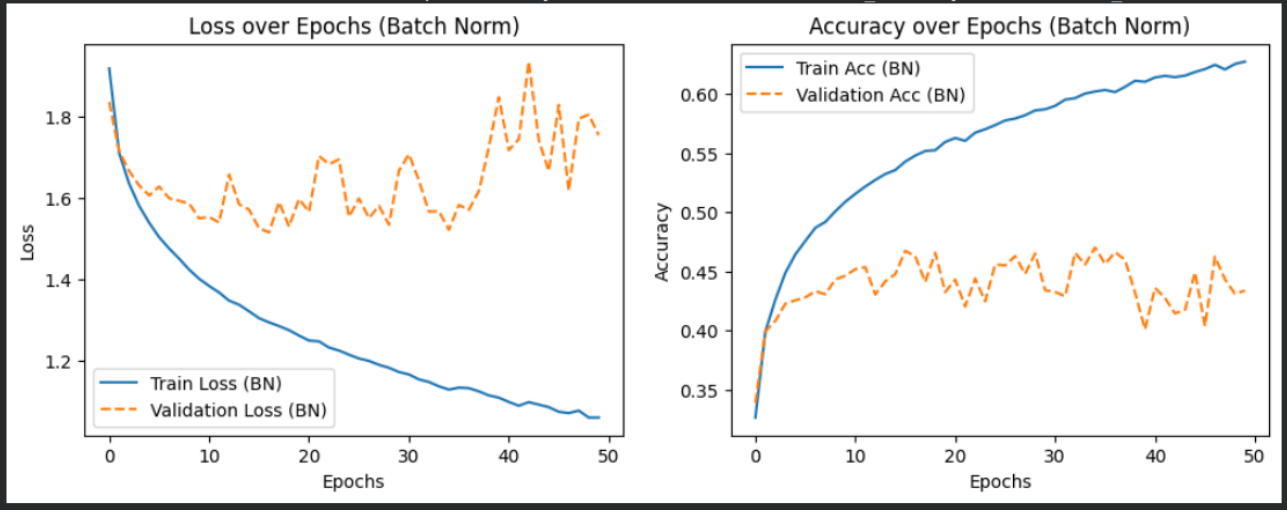


Q7 : Use the early stopping technique



Q8 : Add a batch normalization layer

```
313/313 ━━━━━━━━━━━━━━━━━━━━ 3s 11ms/step - accuracy: 0.6254 - loss: 1.0604 - val_accuracy: 0.4310 - val_loss: 1.8052
Epoch 50/50
313/313 ━━━━━━━━━━━━━━━━━━━━ 3s 9ms/step - accuracy: 0.6274 - loss: 1.0604 - val_accuracy: 0.4336 - val_loss: 1.7550
```



Q9 : Use random search to find the best hyperparameters for learning_rate,dropout_rate, and batch_size,

```
Starting Random Search for 5 combinations...

--- Trial 1/5 ---
Testing: LR=0.01, Dropout=0.2, Batch=128
Result -> Best Validation Accuracy: 0.4641

--- Trial 2/5 ---
Testing: LR=0.001, Dropout=0.5, Batch=64
Result -> Best Validation Accuracy: 0.4257

--- Trial 3/5 ---
Testing: LR=0.001, Dropout=0.5, Batch=32
Result -> Best Validation Accuracy: 0.4311

--- Trial 4/5 ---
Testing: LR=0.1, Dropout=0.3, Batch=128
Result -> Best Validation Accuracy: 0.4103

--- Trial 5/5 ---
Testing: LR=0.01, Dropout=0.3, Batch=128
Result -> Best Validation Accuracy: 0.4560

==================================================
🏆 BEST PARAMETERS FOUND:
Learning Rate: 0.01
Dropout Rate:  0.2
Batch Size:    128
Best Val Acc:  0.4641
==================================================
```

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